

PHASE 3 – PROJECT DESIGN PHASE

Project Title: FitBuddy – AI Fitness Plan Generator using Gemini Models

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1. Design Overview

FitBuddy is designed as a simple AI-assisted web application. The design converts the requirements identified in Phase 2 into a user input flow, Gemini processing flow and structured fitness-plan output.

2. System Architecture

Component	Description
User Interface	Collects age, fitness goal, activity level, workout days and optional preferences.
Application Logic	Validates inputs, prepares the prompt and handles the request/response flow.
Gemini Model	Processes the user information and generates a personalized structured fitness plan.
Output Module	Displays the generated plan in a clear, readable format.
Configuration	Keeps API credentials/private configuration separate from public source code.

3. Proposed Workflow

- User opens the FitBuddy application.
- User enters the required fitness information.
- The application performs basic input validation.
- The entered information is converted into a structured prompt.
- The prompt is sent to the selected Gemini model.
- Gemini returns a generated fitness plan.
- The application displays the plan to the user.

4. User Interface Design

- Input section for age.
- Fitness-goal selection/input.
- Activity-level selection.
- Workout-days selection.
- Optional preferences such as available time or exercise preference.
- Generate Plan button.
- Readable result section for the generated plan.

5. Data Flow

User Input → Validation → Prompt Preparation → Gemini API/Model → AI Response → Structured Fitness Plan → User Display.

6. Design Considerations

- The interface should be simple and mobile-friendly where practical.
- Required fields should be validated before an AI request is made.
- API credentials should not be exposed in public source code.
- The generated output should be formatted so that users can easily read the schedule and recommendations.

7. Design Conclusion

The design phase provides the system structure needed to move FitBuddy from requirements into implementation. It defines the main components, workflow, interface elements and data flow.